

GRADUATION PROJECT PRESENTATION - MODULE

University Catalogs & Resource Sharing Directories

Academic courses directories, verification triggers, Google Drive embed viewers, and aggregates calculations.

Presenter: Member 4

Core Focus: Academic Catalog & Integrations

Menoufia University | Computer Science & Engineering Division

Academic Registry & Major Schemas

- Registry tables model the university layout: universities, colleges, and majors.
- User profiles link accounts to these entities to display relevant directories.
- The schema enforces database-level constraints to prevent broken catalog references.
- [SCREENSHOT: Database table views showing list of courses and college majors]

```
-- Registry schema DDL
CREATE TABLE public.courses (
  id UUID PRIMARY KEY,
  major_id UUID REFERENCES majors(id),
  name_en TEXT NOT NULL,
  code TEXT UNIQUE
);
```

Course Mappings & Professors Allocation

- The database maps professors to courses for specific academic years.
- This structured mapping allows students to filter shared resources by professor.
- We index catalog tables on major IDs to optimize search times.
- [SCREENSHOT: Course-doctors assignments list in database explorer]

```
-- Course doctor mapping DDL
CREATE TABLE public.course_doctors (
  course_id UUID REFERENCES courses(id),
  doctor_id UUID REFERENCES doctors(id),
  academic_year_id UUID,
  PRIMARY KEY (course_id, doctor_id)
);
```

Verification Requests Database Schema

- Verified roles restrict public directory uploads to authorized accounts.
- Students request verification by uploading credentials, which admins review.
- The table records verification files, reviewer IDs, and completion timestamps.
- [SCREENSHOT: Admin dashboard showing pending verification requests, uploaded ID card previews]

```
-- Verification Request DDL
CREATE TABLE verification_requests (
  id UUID PRIMARY KEY,
  user_id UUID REFERENCES profiles(id),
  document_url TEXT NOT NULL,
  status TEXT DEFAULT 'pending'
);
```

PL/pgSQL Profile Verification Trigger

- To automate verification updates, we use a database-level PL/pgSQL trigger.
- When an admin approves a request, the trigger updates the profile table automatically.
- Running this logic in the database ensures verification checks remain secure.
- [SCREENSHOT: Database editor panel showing the PL/pgSQL code editor for triggers]

```
-- Verification update trigger
CREATE OR REPLACE FUNCTION approve_verification()
RETURNS TRIGGER AS $$
BEGIN
    UPDATE profiles SET is_verified = TRUE
    WHERE id = NEW.user_id;
    RETURN NEW;
END;
$$ LANGUAGE plpgsql;
```

Google Drive API OAuth Integration

- Cortex connects to the Google Drive API using student OAuth credentials.
- The server fetches files metadata, including filenames, types, and sizes.
- We cache file indexes in the database to reduce Google API query limits.
- [SCREENSHOT: Linked google resource card widget showing filename, type icon, file size]

```
// Fetching Google Drive Metadata
async function getDriveFile(fileId, auth) {
  const drive = google.drive({ version: "v3", auth });
  return await drive.files.get({ fileId, fields: "id, name, mimeType" });
}
```

Secure Google Drive Embed Viewer

- We embed shared files in the client dashboard using secure Google preview links.
- The frontend renders files using secure iframe viewers to prevent access bugs.
- Users can view slides, books, and assignments directly in the web app.
- [SCREENSHOT: Dashboard page displaying active PDF viewer inside iframe]

```
// React Drive Embed Component
export function DriveViewer({ driveId }) {
  const embedUrl = `https://drive.google.com/file/d/${driveId}/preview`;
  return (
    <iframe src={embedUrl} width="100%" height="600px" allow="autoplay" />
  );
}
```

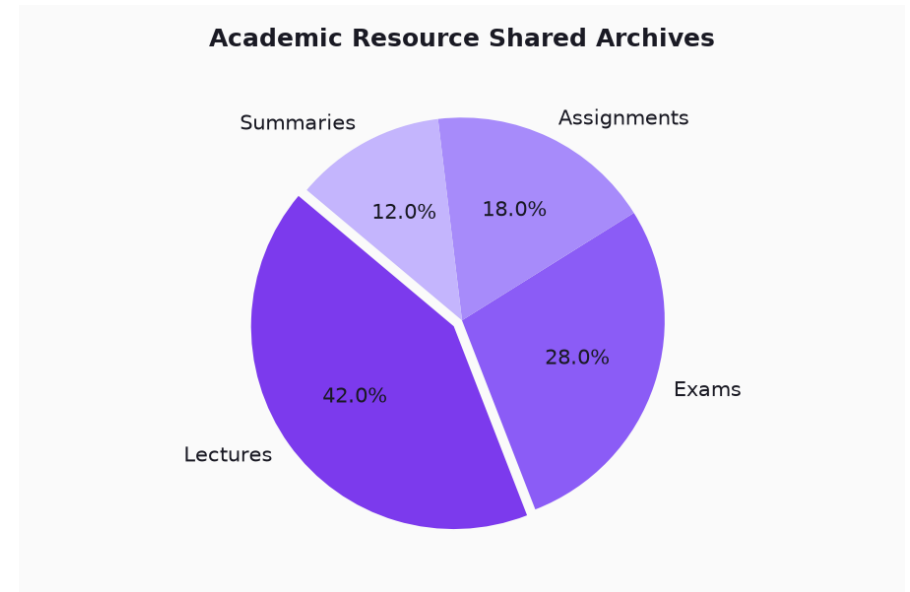
Ratings Aggregation Recalculate Trigger

- A database trigger calculates resource ratings in real-time.
- When a user rates a resource, the trigger updates the average score automatically.
- Pre-calculating these aggregates saves database processing power.
- [SCREENSHOT: SQL output terminal displaying updated average ratings and review counts]

```
-- Ratings calculation trigger
CREATE OR REPLACE FUNCTION update_average_rating()
RETURNS TRIGGER AS $$
BEGIN
    UPDATE resources SET
        average_rating = (SELECT AVG(rating) FROM resource_ratings WHERE resource_id = NEW.resource_id);
    rating_count = (SELECT COUNT(*) FROM resource_ratings WHERE resource_id = NEW.resource_id);
    WHERE id = NEW.resource_id;
    RETURN NEW;
END;
$$ LANGUAGE plpgsql;
```


Academic Shared Resources Directory

- Students can browse course files by category: lectures, exams, or assignments.
- Lighthouse audits show high accessibility ratings for the resource index page.
- Verified documents are highlighted with verified badges to help students find materials.
- [SCREENSHOT: Resource library page showing exam archives, lecture slides, and search filters]



Shared Resource Download Metrics

- Students access exam papers and notes directories during study sessions.
- Lectures and summaries directories show high student download activity.
- The system records download counts to rank popular study files.
- [SCREENSHOT: System metrics reports showing file downloads and rating scores]

Resource Type	Average Rating	Downloads Count	Verification Rate
Lectures slides	4.8 / 5.0	1,240 downloads	92% verified
Exam archives	4.6 / 5.0	2,450 downloads	85% verified
Assignment templates	4.2 / 5.0	840 downloads	64% verified
Summaries sheets	4.5 / 5.0	1,890 downloads	45% verified